

PHARM: TYPE 1 DIABETES MELLITUS & INSULIN

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No conflicts of interest to disclose

Learning Objectives

1. Distinguish the essential pathophysiological differences between type 1 and type 2 diabetes and contrast the role of insulin treatment in these different types of diabetes
2. Explain the pattern of endogenous insulin secretion from the islet cells and how to prescribe exogenously administered insulin to replicate this pattern (basal –bolus concept)
3. Identify the insulin types (basal vs. mealtime) and differentiate their pharmacokinetic profiles
4. Appraise and contrast the differences between human insulins and insulin analogs
5. Describe adverse effects associated with insulin therapy

What is Diabetes Mellitus?

Chronic metabolic disorder characterized by hyperglycemia that results from a variety of pathophysiologic mechanisms

Definitions and Classifications of Diabetes

- Type 1 Diabetes
 - Autoimmune disorder
 - Pancreatic β -cell destruction with essentially complete insulin deficiency
 - Typically presents with acute onset of symptoms
 - 5% of all diabetics / Higher incidence in Caucasians
 - More commonly in young, but can occur at any age

Definitions and Classifications of Diabetes

Type 2 Diabetes: Elevated blood glucose resulting from multiple defects leading to

- insulin resistance (glucose and lipid metabolism) and
- relative insulin deficiency (inadequate β -cell insulin secretion)

Frequently asymptomatic. Develops insidiously over many years

- Intermediate diagnoses – “Pre-Diabetes”
 - Impaired glucose tolerance (IGT)
 - Impaired fasting glucose (IFG)

Secondary Forms of Diabetes: Gestational Diabetes

- Develops during pregnancy and usually subsides after delivery
- Pathogenesis: insulin resistance occurs in all pregnancies
 - As in T2DM, gestational diabetes can occur if β -cell insulin secretion is inadequate
- Incidence of up to ~6% of pregnancies
- Increased fetal morbidity and mortality if not treated
- 30-40% will develop Type 2 DM 5-10 years later
- Women at risk are screened at weeks 24-28

Diabetes Secondary to Other Conditions

- Pancreatic disease – due to destruction of islet tissue (e.g. hemochromatosis, pancreatitis)
- Other endocrinopathies - due to excess hormones that antagonize effect of insulin
 - e.g., cortisol, growth hormone, catecholamines, glucagon
- Drug therapies
 - Corticosteroids – increase insulin resistance

Diabetes Due to Defined Genetic Mutations

- Rarely, mild forms of non-immune mediated diabetes can be associated with a specific genetic defect in proteins known to be important for insulin secretion or insulin action
- A typical example is a rare genetic defect affecting glucose-stimulated insulin secretion, called MODY or “Maturity Onset Diabetes of Youth”

Acute Symptoms of Diabetes

Failure to utilize calories in diet because of a lack of anabolic effect of insulin

Hyperglycemic symptoms:

- Polyuria (increased urination)
- Polydipsia (thirst)
- Dehydration
- Polyphagia (hunger)
- Weight loss
- Blurred vision
- Altered mental status (hyperosmolar states)

Criteria for Diagnosis of Diabetes

A- Fasting glucose ≥ 126 mg/dL OR

B- 2-hr OGTT ≥ 200 mg/dL OR

C- Random plasma glucose ≥ 200 mg/dL
with symptoms OR

D- A1c $\geq 6.5\%$

Criteria A,B & D should be confirmed by repeat testing

Laboratory Diagnosis of Diabetes Mellitus: Oral Glucose Tolerance Testing Criteria

Oral glucose tolerance test (OGTT) criteria

75 gm oral glucose given in fasting state

- Glucose ≥ 200 mg/dL at 2 hr = DIABETES
- Glucose 140-199 mg/dL at 2 hr = Impaired Glucose Tolerance (“Pre-Diabetes”)
- Glucose < 140 mg/dL at 2 hr = Normal Glucose Tolerance

Diagnosis by A1c

Non-enzymatic glycation of hemoglobin; the degree of glycation reflects the average glucose exposure over the half-life of the RBC. A1C is therefore influenced by red cell survival

- Normal: $< 5.7\%$
- Increased risk of diabetes: $5.7-6.4\%$
- Diabetes $\geq 6.5\%$

Also used to monitor control of diabetes

- Should generally be 7.0% or less (American Diabetes Association)

Hemoglobin A1c

(most widely used test to estimate glucose control)

HbA1c	Mean Blood Glucose
6.0%	120 mg/dl
7.0%	150 mg/dl
8.0%	180 mg/dl
9.0%	210 mg/dl
10.0%	240 mg/dl
11.0%	270 mg/dl

Type 1 Diabetes

**Autoimmune destruction of beta cells,
set off by an environmental trigger in a
genetically susceptible person**

Type 1 Diabetes

Genetic Component: Strong HLA association (HLA-DR 3 and DR4 in 90-95% of patients). Multiple genes may be involved (polygenic)

Environmental component: factors/triggers

- ?Viruses / unknown that set off immune cascade

Type 1 Diabetes

Autoimmune (T cell mediated) destruction of β - cells

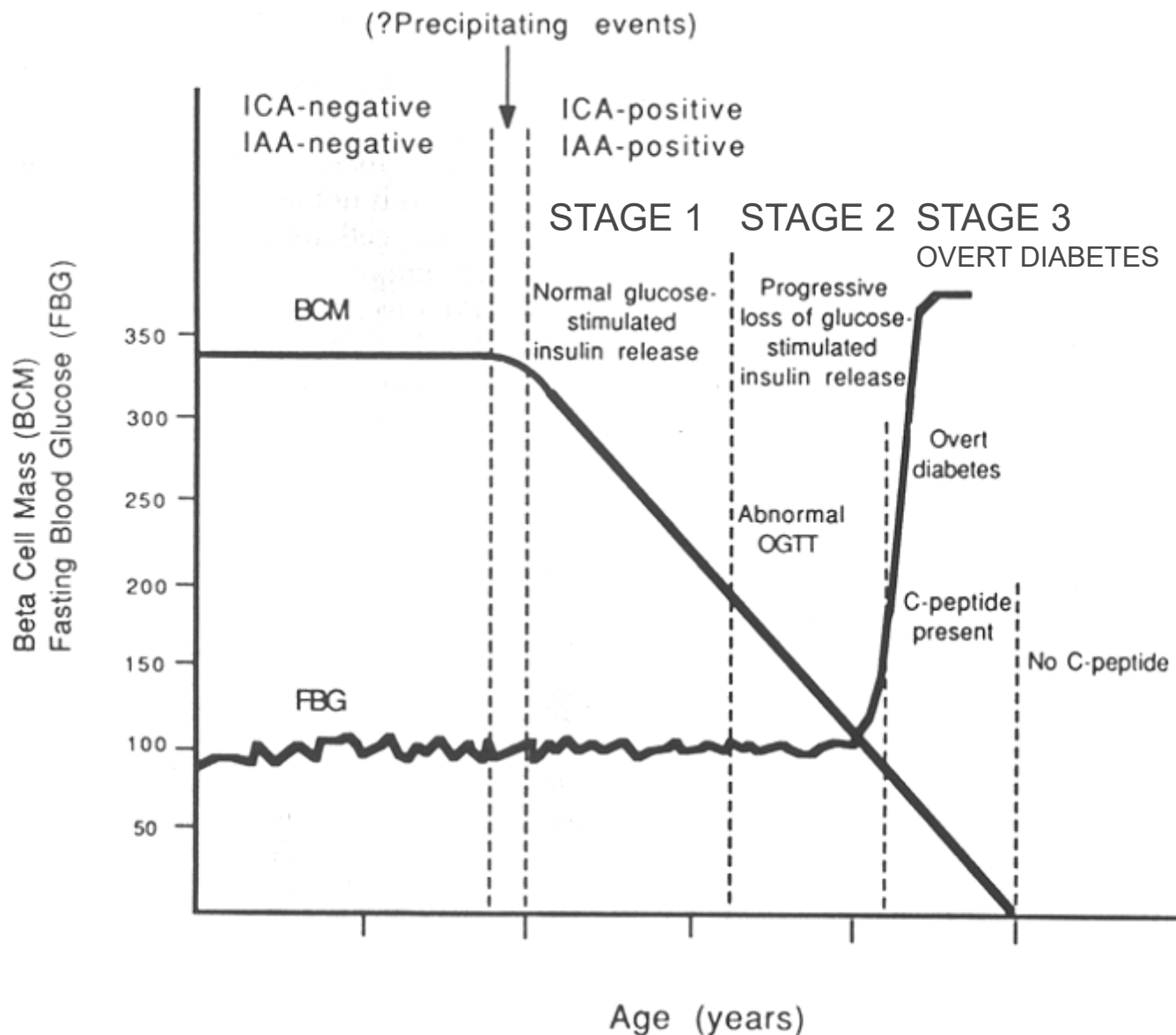
Antibody markers of immune destruction measurable in up to 90% of patients

Islet cell components (GAD65- Glutamic Acid Decarboxylase 65, IA2- tyrosine phosphatases, islet cell antibodies, Zinc transporter 8)

Prone to other autoimmune diseases (thyroid, addisons, vitiligo)

Eventually, > 90% of β -cells are destroyed (hyperglycemia develops with possible ketoacidosis)

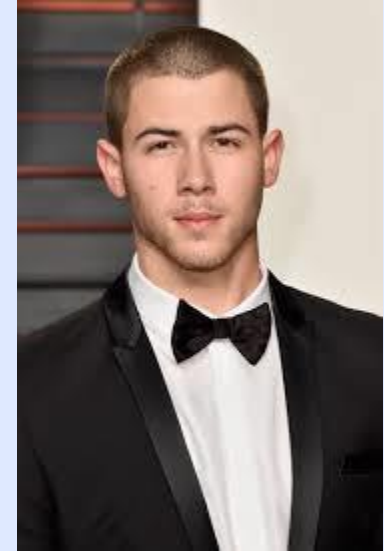
Pathogenesis of Type I Diabetes



Type 1 Diabetes

- Little or no insulin secretion (low or undetectable levels of plasma C-peptide)
- Rate of β - cell destruction variable (determines presentation at onset)
- All become dependent on insulin for survival
 - At risk for Diabetic Ketoacidosis (DKA)

Famous Type 1 Diabetics



Predictability of Type 1 Diabetes

- Risk of developing Type 1 Diabetes:
 - In the general population = 0.3 - 0.5%
 - With one first degree relative = 2-6%
 - In monozygotic twins = 30-40%;
 - up to 70% in twins with HLA haplotypes DR3 / DR4
 - If mother has Type 1 diabetes = 2-3%
 - If father is affected = 5-6%

Prevention of Type 1 Diabetes

- Large trials have tested whether progression to Type 1 diabetes could be prevented
 - Individuals at high risk
 - Usually first-degree relatives with (+) antibodies
- Randomized to various agents
 - Low dose insulin injections - ? Tolerizing effect to antigen
 - Immune modulators

Prevention of Type 1 Diabetes?

TEPLIZUMAB (Tziel)

First agent approved (in 2022) to **DELAY (not prevent)** the onset of Type 1 diabetes by up to 2 years

Mechanism of action: binds to CD3 on T cells, to prevent signaling that leads immune destruction of the beta cells

- Indications: stage 2 diabetes- at least 2 antibodies and dysglycemia, but not overt diabetes
- Infusion daily over 14 days
- High cost
- Side effects- lymphopenia (monitor WBC during treatment), infection

“Typical” Presentation of Type 1 Diabetes

Thin younger person

Polyuria, polydipsia, weight loss, fatigue, blurred vision

Can be dramatic with DKA (Diabetic Ketoacidosis)

- Acetone odor on breath (some call “fruity”)
- Insulin deficiency leads to ketone formation
- Nausea, vomiting, metabolic acidosis
- Up to 10% mortality- a medical emergency
- Must be treated with insulin, fluids

Can draw blood for antibodies to confirm dx

Type 1 vs Type 2 Diabetes

	TYPE 1	TYPE 2
Out-Dated names	“Insulin-Dependent” Juvenile Diabetes	“Non-Insulin Dependent” Adult-Onset
Out-Dated acronyms	IDDM	NIDDM
Age of onset (most often)	Usually younger	Usually adults
Ketosis	Present	Absent
Recent Weight Loss	Yes	Usually no
Time course of onset	Abrupt with symptoms	Prolonged (over years) typically with no symptoms
Overweight/obese	Not usually	Most of the time

Type 1 vs Type 2 Diabetes

	TYPE 1	TYPE 2
Insulin Resistance	No	Yes
Islet Cell Antibodies & GAD-65 Antibodies	Yes	No
Insulin Secretion Defect	Yes	Yes
HLA-Associated Antigens	Yes	No
Associated with other Auto-immune Diseases	Yes	No
Insulin Rx Necessary	Absolutely	Sometimes

TYPE 1 DIABETES

QUESTIONS?

INSULIN

Who Needs Insulin Therapy?

- Type 1 Diabetes - ALL patients need
 - Insulin always to prevent ketoacidosis
- Type 2 Diabetes
 - Hyperglycemia despite maximally effective doses of oral or non-insulin injectable agents
 - Significant hyperglycemia at presentation
 - Acute sickness, Hospital stay, Surgery
 - Pregnancy

Clinical Use of Insulin

Advantages

- Can control all patients
- Flexibility in dosing
- Pen devices improve compliance
- Newer long and short-acting insulins provide clinical benefit

Disadvantages

- Injections required
- Hypoglycemia
- Weight gain
- Can be expensive (worse if no insurance)

Endogenous Insulin

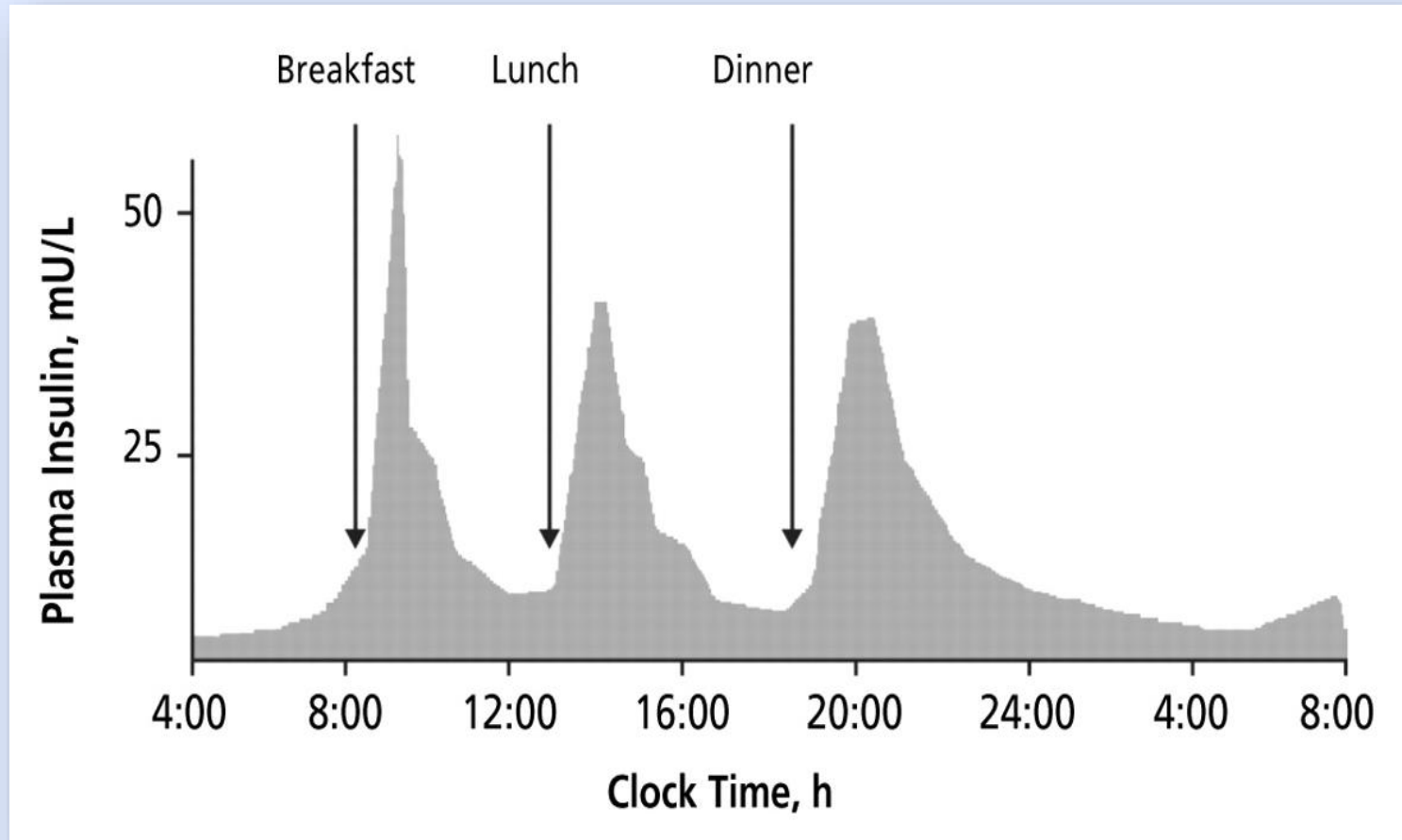
Normal physiology

Continuous release of insulin throughout the day ('basal insulin') regulates lipolysis and the output of hepatic glucose

Spikes of insulin release in response to a meal ('boluses').

- Occurs in a robust first-phase secretion and then a more prolonged second-phase release into the portal circulation

Physiologic Serum Insulin Secretion Profile



Types of Insulin

- 1921
Discovery of Insulin (Animal insulins- porcine, bovine)
- 1980's
Purification of animal insulin, recombinant Human insulins:
Regular NPH (Lente- off market)
- 1990's and beyond
Analog insulin developed: Human insulin changed by recombinant DNA techniques -reversing or substituting amino acids in human insulin:
Lispro (Humalog®, Lyumjev®) Glargine (Lantus®)
Aspart (Novolog®, Fiasp®) Degludec (Tresiba®)
Glulisine (Apidra®)
(Note Detemir (Levemir®) discontinued in 2024)

Types of Insulin

- Insulins divided based on duration of action

Rapid

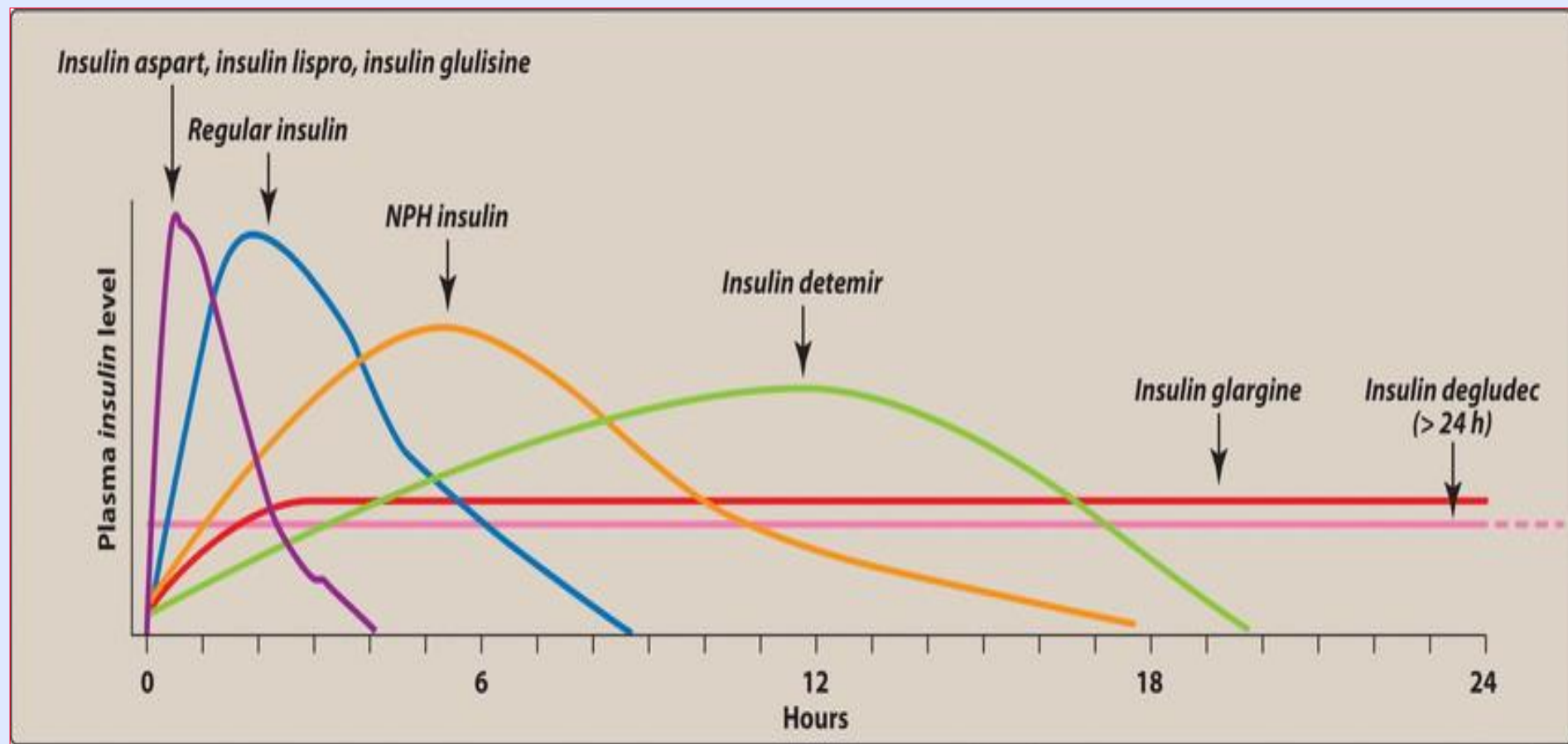
Short

Intermediate

Long

Human and Analog Insulins

Type	Name	Onset	Peak	Duration
Rapid-acting analogs (bolus insulins)	lispro, aspart, glulisine	15 min	1-2 h	3-5 h
Short-acting (old bolus insulin)	regular	30-60 min	2-4 h	6-8 h
Intermediate-acting (old basal insulin)	neutral protamine Hagedorn (NPH)	2-4 h	4-10 h	12-16 h
Basal insulin analogs	glargine	2-3 h	(peakless)	20-24 h
	degludec	1h	(peakless)	≥ 24 h

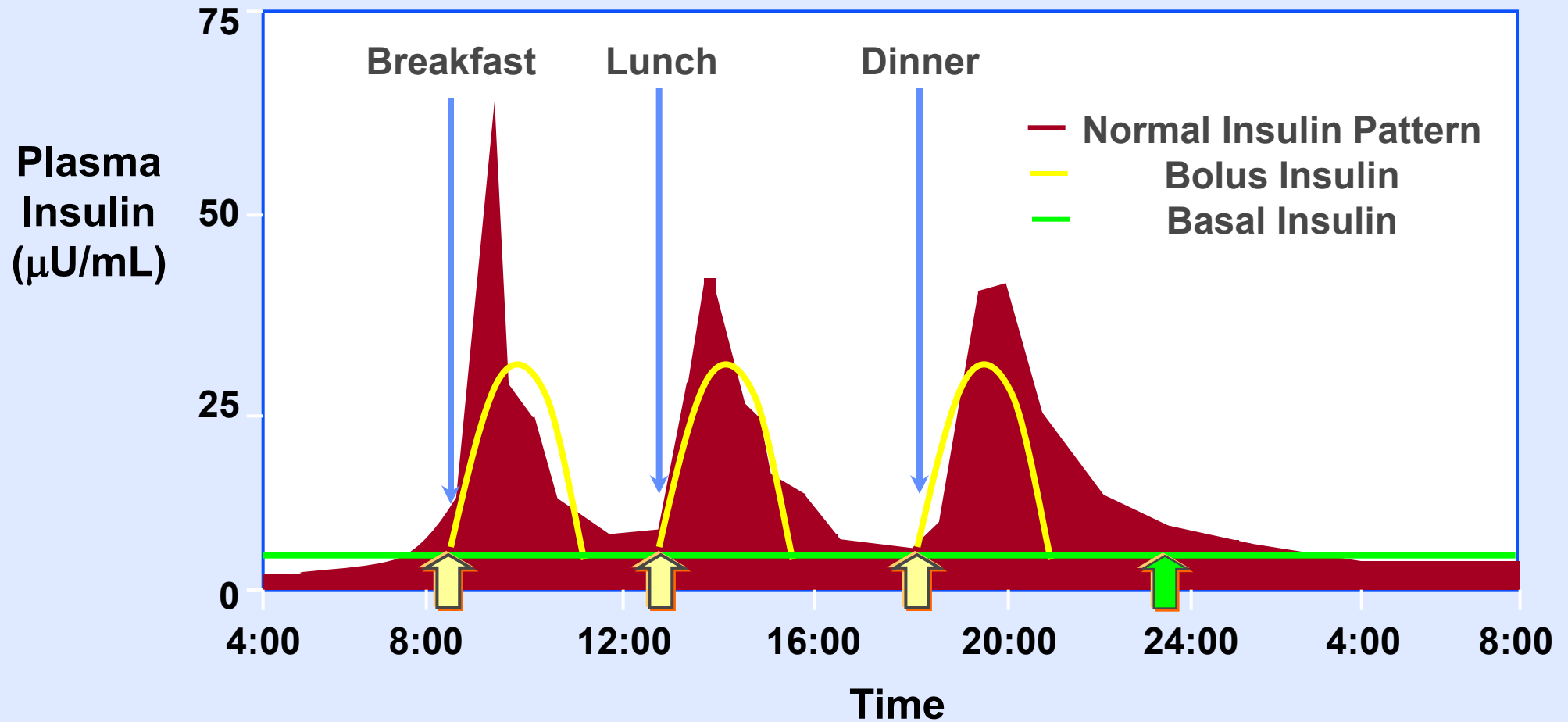


Basal- Bolus

Using insulin injections to try to mimic the 'normal' pancreas- a more physiologic regimen.

- “Basal” is intermediate or long acting insulin
 - Suppresses hepatic glucose output and ketogenesis
 - Approximately 50% of total insulin requirement
- “Bolus” is the short or rapid acting insulin prior to the meal
 - Faster delivery into bloodstream as food is absorbed

Ideal Basal-Bolus Insulin Absorption Pattern



Basal Insulin (Analogues)

Insulin glargine (Lantus®, Basaglar®), detemir (Levemir®), degludec (Tresiba®)

- Structurally-altered insulin analogs
 - glargine - soluble at acid pH, but forms micro-precipitates in subcutaneous space that retard absorption
 - degludec forms multi-hexamers when injected into the subcutaneous tissue resulting in a subcutaneous insulin depot
- All 3 are “peakless” compared to the older insulin NPH:
 - Less weight gain
 - Less hypoglycemia

Basal Insulin (Analog)

Insulin Icodec - abae (Awiqli®)

First FDA approved ONCE WEEKLY insulin (2026)

Reversibly binds to albumin via diacid side chain and slowly dissociates

Rapid Acting Insulin (Analogues)

- Insulin lispro (Humalog®), aspart (Novolog®), glulisine (Apidra®)
- Structural analogues of insulin that facilitate dissociation from Zn crystal
- Faster onset of action
 - Limits postprandial hyperglycemic peaks
- Shorter duration of activity
 - Minimizes late postprandial *hypoglycemia*
- Convenient administration immediately prior to meals

Rapid Acting Insulin (Analog)

Even faster acting analogs- additives to increase absorption:

- Insulin lispro aabc (Lyumjev®)- prostacyclin (trepostinil) and citrate to increase local vascular permeability and dilation)
- Insulin aspart (Fiasp®)- vitamin B3 (niacinamide) which causes a transient local vasodilation

Administration immediately prior to meals or during/ after meal (but **preferred** to dose pre-meal)

Intermediate Acting NPH Human Insulin

Older insulin (used before basal analogs)

Crystallized human insulin with protamine and zinc

Does not mimic basal profile

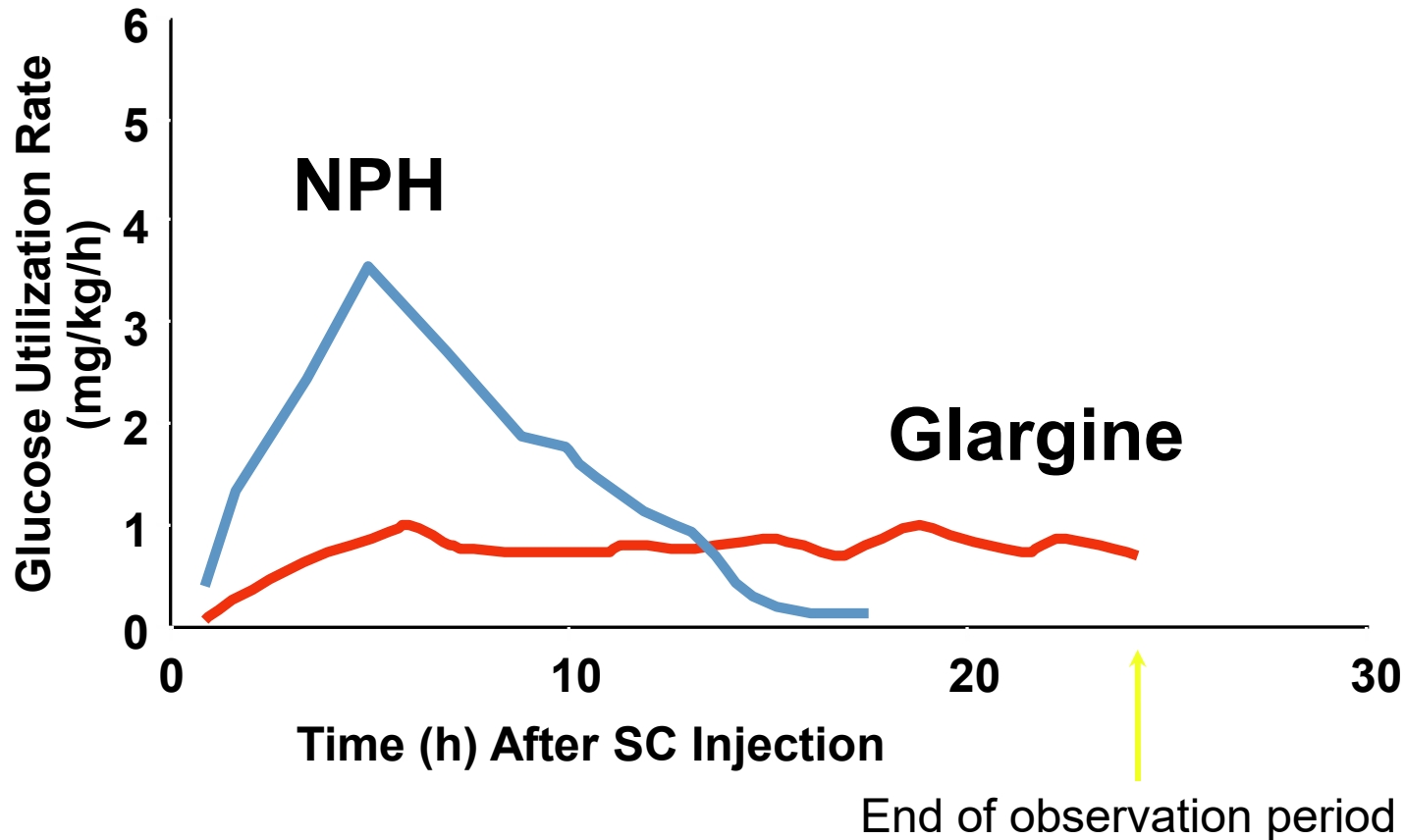
- Pronounced peaks
- Less than 24 hour duration

Can cause unpredictable hypoglycemia

- Which will limit insulin adjustments
- Can lead to weight gain

Glargine vs NPH Insulin

Action Profiles by Glucose Clamp



Lepore, et al. Diabetes. 1999;48(suppl 1):A97.

Short Acting Regular Insulin

Human Insulin

Older insulin (used before rapid acting analogs)

Crystalline soluble zinc insulin

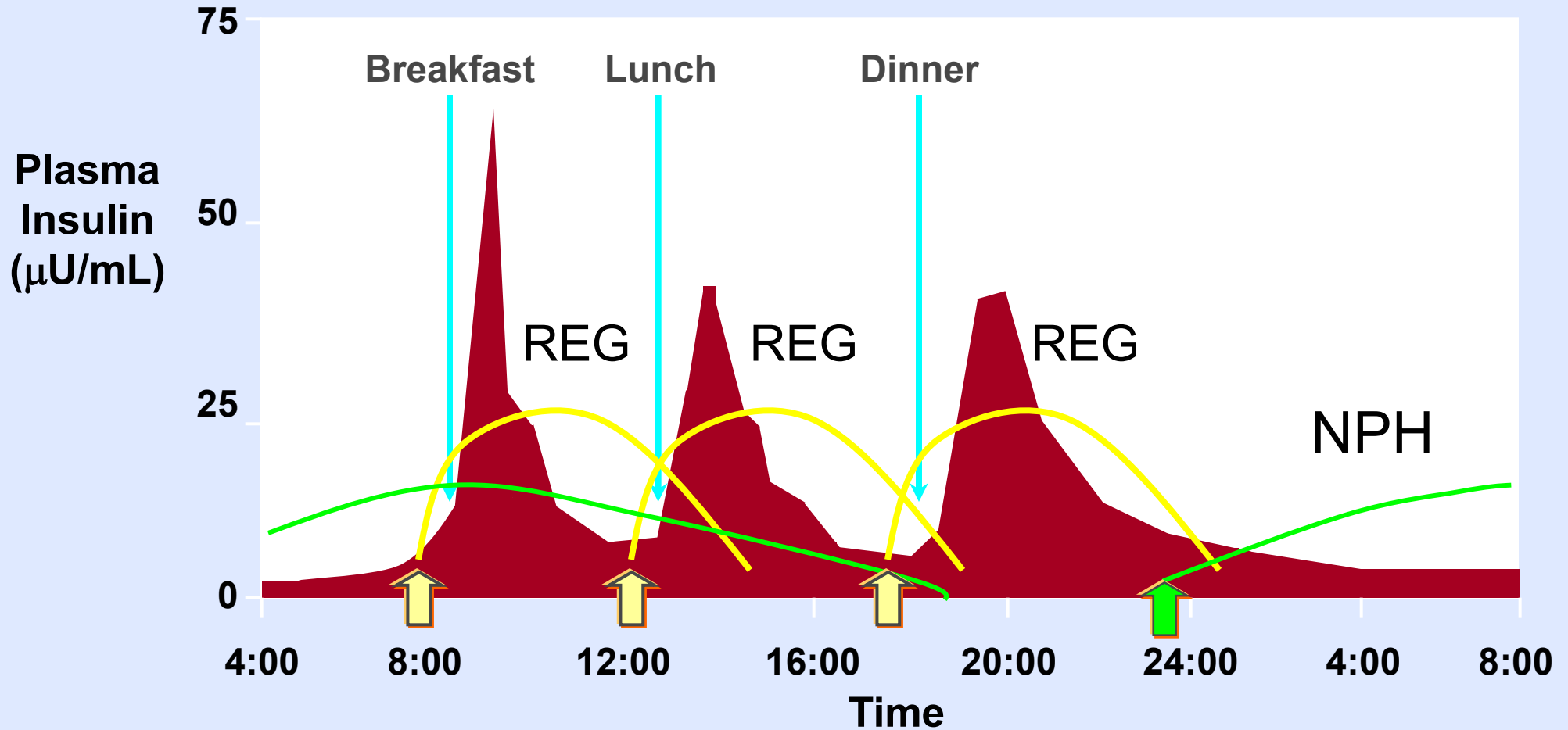
Slower onset of action, inconvenient administration (30 to 45 minutes prior to meal)

- Risk of hypoglycemia if meal is delayed
- Mismatch with postprandial hyperglycemic peak

Up to 6-8 hours duration

- Potential for late postprandial hypoglycemia

Limitations of Regular + NPH Insulin



We Still Use Human Insulins

Cost is lower (if no prescription coverage, if high copays)

Time action profile may match other circumstances (ex. NPH may target late day hyperglycemia in patients on morning steroids)

Basal-Bolus Analog Regimens are not Perfect

Using FIXED amounts of mealtime insulin is not as physiologic

Ex. A patient is on insulin glargine 30 units daily and aspart 5 units with each meal

Monday morning the patient ate cereal and fruit (40 carbs)

Tuesday morning the patient ate eggs and bacon (0 carbs)

Must assess patient: Compliance (4 shots per day), provide education

- BEST way to do this is Carbohydrate counting
How many units of insulin will cover the amount of carbs ingested

Premixed Insulin Formulations

- Not all patients are willing to inject multiple times daily. Premixed insulins (less optimal) may offer convenience (2 shots per day)
- Combines rapid or short-acting insulin with an intermediate-acting insulin
- For example:
 - “70/30” = 70 Unit/ml NPH + 30 Unit/ml Regular
 - Novolog “70/30” = 70 Unit/ml aspart/protamine complex + 30 Unit/ml aspart
 - Humalog “75/25” = 75 Unit/ml lispro/protamine complex + 25 Unit/ml lispro
 - Humalog “50/50” = 50 Unit/ml lispro protamine complex + 50 Unit/ml lispro

Concentrated insulins

- Patients on large doses of insulin
- Can get same amount of units, but in a lower volume
 - R U-500 Insulin
 - Glargine U-300
 - Humalog U-200

Inhaled insulin

Inhaled human insulin (Afrezza®): Ultra-rapid acting human insulin with immediate onset, peak in 15 minutes and duration of 3 hours

Not commonly used

Contraindications (smokers, chronic lung disease)
Need to monitor pulmonary functions tests

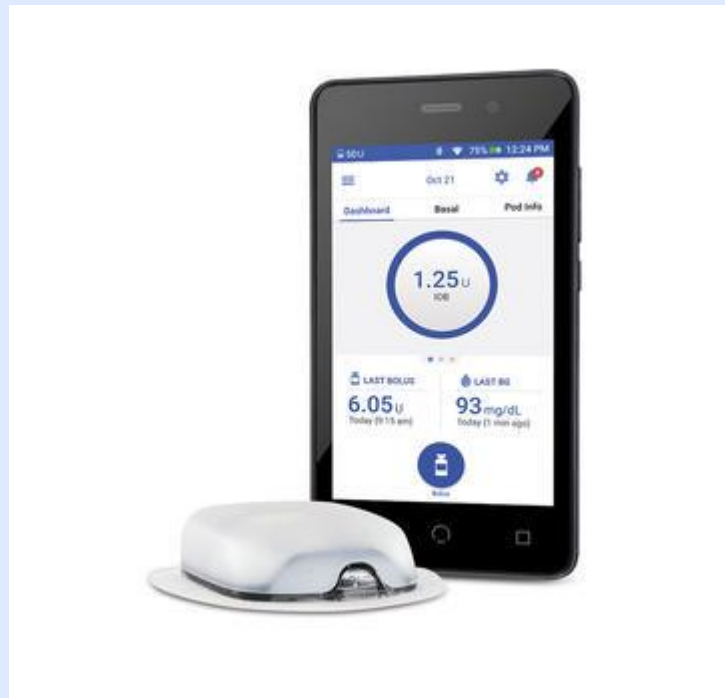
Most need repeated loading of cartridges to make one mealtime dose

Insulin Pumps

CSII –Continuous Subcutaneous Insulin Infusion

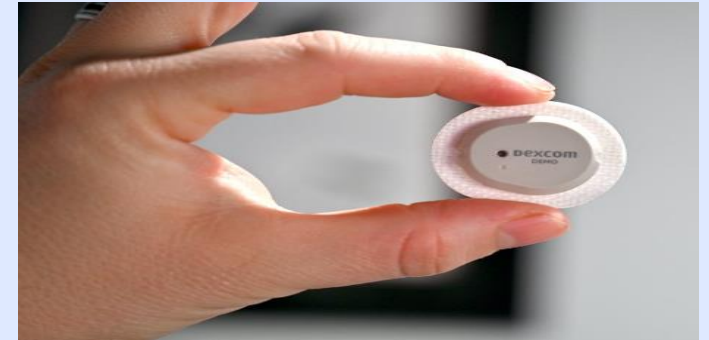
- Insulin infused by mechanical pump, programmable
- Attached through a *subcutaneous catheter*
- Rapid acting insulin infused continuously (basal insulin)
- Patient boluses rapid acting insulin prior to meals or to correct high sugars (bolus insulin)
- Disadvantages- cost, being attached to a device





Continuous Glucose Monitoring

- Devices that measure interstitial fluid glucose (subcutaneous)
 - Lags behind capillary glucose
 - Can see glucose trends
 - Important tool for managing type 1
 - Cost / accessibility – not available to everyone
-
- Newer automated insulin delivery systems: continuous glucose monitor information goes to insulin pump and algorithm helps with insulin delivery (not an artificial pancreas)



AGP Report

Name _____

MRN _____

GLUCOSE STATISTICS

26 Feb 2019–10 Mar 2019
% Time CGM is Active

13 days
99.9%

Average Glucose
Glucose Management Indicator (GMI)
Glucose Variability

173 mg/dL
7.6%
49.5%

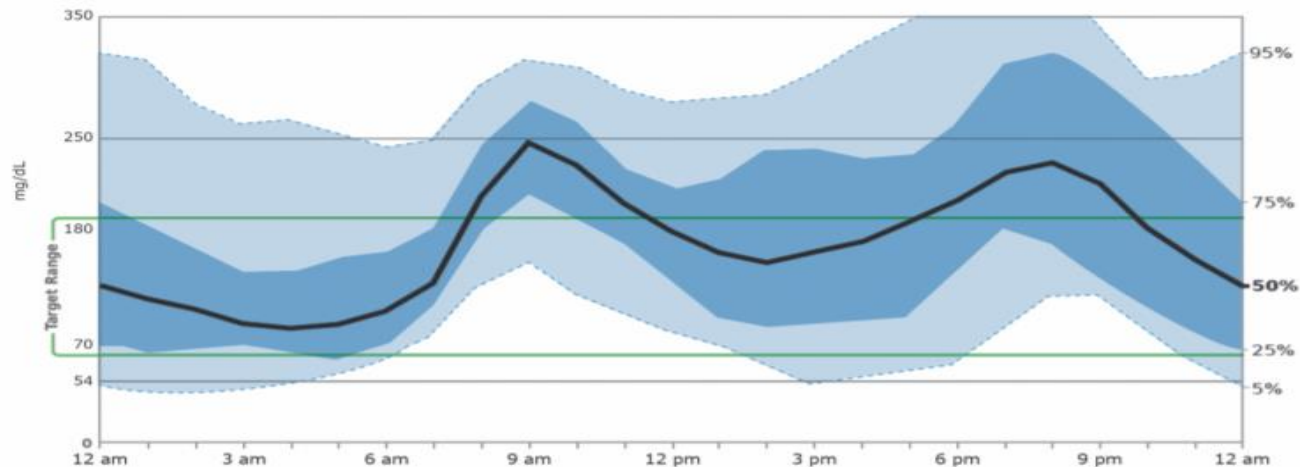
Defined as percent coefficient of variation (%CV); target $\leq 36\%$

TIME IN RANGES



AMBULATORY GLUCOSE PROFILE (AGP)

AGP is a summary of glucose values from the report period, with median (50%) and other percentiles shown as if occurring in a single day.



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QUESTIONS?